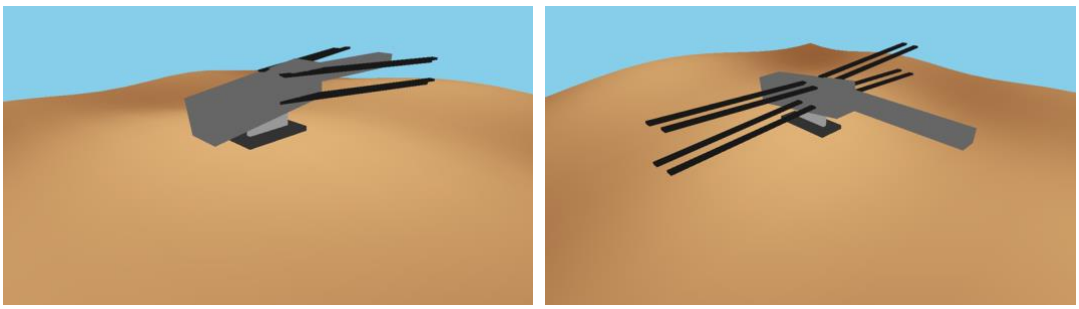
Proseminar Visual Computing
Winter Semester 2026 / 2027

Assignment 3

Hand-out: 10th Nov 2026 **Hand-in: 23rd Nov 2026**



Topics

- General OpenGL programming
- Transformations
- Basic animations and user controls
- Camera control

Outline

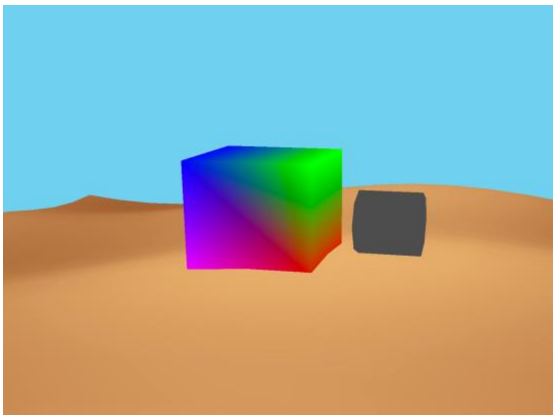
The goal of the Computer Graphics assignments of the Visual Computing PS is to build an animated scout ornithopter from Dune. This work is divided into 3 steps. Each step corresponds to a programming assignment. In this first assignment, we focus on geometric transformations and animation. The objective of this first assignment is to build a very simple ornithopter made from basic geometric primitives by applying different transformations.

Further, the ornithopter should be controlled using user input and when flying around, the wings should be animated. Finally, an additional camera mode should be added that follows the ornithopter. See the example video provided with this assignment for a working solution including all features. You can also check out this video to see the wing opening and flapping animation of the ornithopter you must recreate.

Template code

A template code is provided for this assignment. You can modify this code to build your own scene. Implementation can mainly be done in the main source file assignment_3.cpp, but feel free to create your own files (e.g., it might be helpful to use a ornithopter.cpp/h) and implement it wherever you want.

The scene available in the template code contains a wavy desert terrain, and two cubes that can be rotated around its *x*- and its *y*-axis using the W, A, S, D keys. Moreover, an orbit camera with the cube in its center is available. The camera can be controlled by holding the left mouse button; it follows the mouse movements and rotates on a sphere with a fixed radius. You can zoom in and out (*i.e.* de- and increase the radius of the sphere) with the mouse wheel. In the figure below, an example of the template scene is depicted:



Tasks

1. **Set up the hierarchical geometrical model of the ornithopter.** For that, disable the scaled cube model from the template code. Then start with 12 cubes (size: 2×2×2) and apply transformations to them (scaling, rotation, and translation) to obtain the ornithopter parts representing: cockpit, tail boom, support strut, landing plate, and 8 identical wings (see introduction figure). All the necessary measurements for the different parts can be taken from the drawing in *dune_ornithopter_drawing.pdf*.

2. **Add user control and animation to the ornithopter.** It should be possible to fly the ornithopter around. In addition, the wings should flap and the ornithopter should be animated while flying.

    - In a first step, the wings should slowly open from their rest position until they are perpendicular to the ornithopter's body, then flip 180° around X so they are right-side up.

    - Next, the wings should start flapping and should speed up over time. You can use the helper function provided for this part.

    - It should be possible to control the ornithopter's forward and backward movement by pressing W and S, respectively. As animation during flying forward/backward, the ornithopter should tilt forward/backward along its pitch axis.

    - Like forward and backward controls, flying left and right should be controllable by pressing A and D. For the animation, tilting along the ornithopter roll axis should be implemented.

    - Flying up and down should be controllable too (e.g. using the Left Shift and the Space key).

    - Likewise, it should be possible to rotate the ornithopter around its yaw axis counter- and clockwise by pressing Q or E.

3. **Finally, implement a second camera mode.** In the second mode the camera should remain stationary but always point towards the ornithopter. Use two different keys (*e.g.*, 1 and 2) to switch between the two modes. If switching from mode 2 to mode 1, it is not required to reset the focus to the origin but just keep the last focus point from mode 2.

Implementation Remarks

Make sure that your code is clear and readable. Write comments if necessary. Your solution should contain a readme file with the names of the team members, a list of keyboard controls (if they are different from suggested), and any explanation that you think is necessary for the comprehension of the code.

Submission and Grading

Submission of your solution is due on November 23rd, 2026 (23:59). Submit the sources (*i.e.*, only the content of the *src* folder) in a ZIP archive via OLAT. Do not submit the executable and the content of the *build* folder. Do not submit the external dependencies either. Both folder and archive should be named according to the following convention:

Folder: **CGA3_<lastname1>_<lastname2>_<lastname3>**

Archive: **CGA3_<lastname1>_<lastname2>_<lastname3>.zip**,

with <lastname1>, etc. the family names of the team members. Development in teams of two or three students is requested. Please respect the academic honor code. In total there are 14 marks achievable in this assignment distributed as follows:

- Geometrical model (**3 marks**)
- Ornithopter control and animations
  - Wing opening and flapping animation (**4 marks**)
  - Ornithopter flying controls and animation (**4 marks**)
- Additional camera mode (**1 marks**)
- Code readability, comments, and proper submission: (**2 marks**)

Resources

- Lecture, proseminar slides, assignment description video, and the template code are available via OLAT.
- OpenGL homepage
  http://www.opengl.org
- OpenGL 3.3 reference pages
  https://www.khronos.org/registry/OpenGL/specs/gl/glspec33.core.pdf OpenGL tutorial
  https://learnopengl.com/
  http://www.opengl-tutorial.org GL framework GLFW
  https://www.glfw.org/documentation.html

Note: Be mindful of employed OpenGL and GLSL versions!